

## CUT&Tag FAQ

### **Q1: Is it feasible to send fixed cells for CUT&Tag?**

A1: It is not recommended. Reasons: 1. The condition of the cell samples before fixation is unknown to us; 2. The effect of cell fixation cannot be quality controlled (a. The concentration of the fixative may be too low to properly fix the chromatin state; b. The concentration of the fixative may be too high, causing the chromatin to shrink, which can result in enzymes not being able to cut effectively during subsequent library construction experiments, etc.).

### **Q2: Why should tissue samples not be preserved with RNAlater, TRIzol, or alcohol?**

A2: These reagents can react with subsequent lysis reagents, affecting the experimental outcomes. For instance, RNAlater can affect the structure of DNA and proteins, hence it is not recommended to use these types of reagents for sample treatment.

### **Q3: What should be noted when performing CUT&Tag analysis?**

A3: It is essential to have a reference genome with complete annotations; for combined analyses, it is important to ensure that the genomic references are consistent and that the sample naming conventions are uniform.

### **Q4: Can CUT&Tag data be analyzed together with data from traditional sonication-based ChIP sequencing?**

A4: It is not recommended because the library preparation methods are different.

### **Q5: What is the difference between ATAC-Seq and CUT&Tag?**

A5: CUT&Tag is used when there is a specific transcription factor of interest prior to the experiment. Antibodies targeting this transcription factor are designed and used in

the CUT&Tag experiment to pull down DNA, thereby verifying whether the transcription factor of interest interacts with DNA. In contrast, ATAC-Seq does not focus on a specific transcription factor. It detects the openness of chromatin across the entire genome, providing information on potential binding sites genome-wide. This technique can be combined with other methods to screen for regulatory factors of interest.